FLIGHT CONTROLLER

ARK FPV

STM32H743 flight controller for 30.5 mm stacks, powered directly from 3S–12S

The USA-built ARK FPV puts an STM32H743 processor, IMU, barometer, magnetometer and 1 W sensor heater on a 30.5 mm stack board. It runs directly from a 3S–12S battery, with a 5 V 2 A regulator shared by the board and peripherals and a switched 12 V 2 A rail for video transmitters and payloads.

It supports PX4, ArduPilot and Betaflight. PX4 ships by default.



KEY FEATURES

- STM32H743IIK6, 480 MHz, 2 MB flash, 1 MB RAM
- TDK InvenSense IIM-42653 industrial IMU; ST LSM6DSV32X on the ARK FPV 32X
- Bosch BMP390 barometer and ST IIS2MDC magnetometer
- Direct 3S–12S battery input (5.5–54 V)
- 5 V 2 A and 12 V 2 A regulators; 12 V switched by the flight controller
- 9 PWM outputs, DShot and bidirectional DShot capable
- 1 W sensor heater under closed-loop control
- microSD logging on a 4-bit SDMMC bus

SPECIFICATIONS

PROCESSOR & MEMORY	
MCU	ST STM32H743IIK6, 480 MHz
Flash / RAM	2 MB / 1 MB

ELECTRICAL	
Battery input	5.5–54 V (3S–12S)
Input clamp	54 V standoff, 60–66.3 V breakdown
5 V regulator	2 A, including the board's own draw
12 V regulator	2 A, switched; follows VBAT below 12 V
Board consumption	500 mA at 5 V, including 200 mA heater
PWM outputs	3.3 V logic, 220 Ω series, signal only
Voltage monitoring	Battery, 5 V, 12 V, 3.3 V sensors

ENVIRONMENTAL & MECHANICAL	
Operating temperature	–25 °C to +85 °C
Storage temperature	–25 °C to +85 °C
Dimensions	36 × 36 × 8 mm
Weight	7.5 g with microSD card
Mounting	4 × Ø3.8 mm on 30.5 × 30.5 mm
Included	microSD card, 4 vibration dampers

AT A GLANCE	
MCU	STM32H743IIK6
IMU	IIM-42653 / LSM6DSV32X
Baro / Mag	BMP390 / IIS2MDC
Input	3S–12S, 5.5–54 V
Regulators	5 V 2 A, 12 V 2 A
Operating temp	–25 to +85 °C
Dimensions	36 × 36 × 8 mm
Weight	7.5 g
Hardware rev	Rev 4.2

COMPLIANCE & SOURCING

- Designed and manufactured in the USA
- NDAA compliant
- Blue UAS Framework listed (ARK FPV)

ORDERING & RESOURCES

STORE arkelectron.com/product-category/flight-controllers/fpv/

DOCS docs.arkelectron.com/products/flight-controller/ark-fpv

CAD github.com/ARK-Electronics/ark-hardware

VARIANTS

	ARK FPV	ARK FPV 32X
SKU	ARK-FPV	ARK-FPV-32X
IMU	TDK InvenSense IIM-42653	ST LSM6DSV32X
Hardware revision ID	ARKFPV000	ARKFPV001

Same board, connectors and pinout; both IMUs reach ±32 g and ±4000 °/s. Blue UAS Framework listing covers the ARK FPV only.

SENSOR BUS MAP

DEVICE	INTERFACE	ADDRESS / SELECT	FUNCTION
IIM-42653 or LSM6DSV32X	SPI1	nCS1, DRDY1	6-axis accelerometer / gyroscope
BMP390	I2C2	0x76	Barometric pressure
IIS2MDC	I2C4	0x1E	3-axis magnetometer
GPS connector	I2C1	External	Compass on a GPS module
SPI connector	SPI6	nCS1, DRDY1, nRESET	OSD or external IMU

SERIAL PORT MAPPING

UART	CONNECTOR	PX4	ARDUPILOT
USART1	GPS	GPS1, /dev/ttyS0	SERIAL3
USART2	VTX (RX only)	TELEM3, /dev/ttyS1	SERIAL4
USART3	DEBUG	Console, /dev/ttyS2	—
UART4	ESC (RX only)	TELEM4, /dev/ttyS3	SERIAL5
UART5	VTX	TELEM2, /dev/ttyS4	SERIAL2
USART6	RC	RC, /dev/ttyS5	SERIAL6
UART7	TELEM (RTS/CTS)	TELEM1, /dev/ttyS6	SERIAL1

FIRMWARE

FIRMWARE	BUILD TARGET	NOTES
PX4	`ark_fpv_default`	Multicopter. Ships installed
PX4	`ark_fpv_fw, ark_fpv_vtol`	Fixed-wing, VTOL
ArduPilot	`ARK_FPV`	—
Betaflight	`ARK_FPV`	—

NOTES

- PWM outputs are 3.3 V logic straight from the MCU's timers through 220 Ω series resistors, so each sources at most ~15 mA, even into a short. The PWM connectors carry no servo power.
- PX4 runs DShot and bidirectional DShot on outputs 1–8; output 9 is PWM only. ArduPilot runs DShot on all 9 and bidirectional DShot on outputs 1–4.
- The ESC connector's pin order matches the ARK 4IN1 ESC, which ships with the cable. CURRENT_IN reaches the ADC through 220 Ω with no divider, so the current scale is the ESC's own.
- The flight controller switches the 12 V rail (pit switch); with the battery below 12 V it follows the battery. USB-C alone powers the board and its 5 V outputs; the 12 V rail needs the battery.
- CAN1 is terminated on the board (120 Ω).

CONNECTOR PINOUTS

ESC · 8-PIN JST-GH

PIN	SIGNAL	VOLTAGE
1	VBAT_IN	5.5–54 V
2	CURRENT_IN	0–3.3 V
3	UART4_RX	3.3 V
4	PWM1	3.3 V
5	PWM2	3.3 V
6	PWM3	3.3 V
7	PWM4	3.3 V
8	GND	GND

SPI (OSD OR IMU) · 8-PIN JST-SH

PIN	SIGNAL	VOLTAGE
1	5V_OUT	5 V
2	SPI6_SCK	3.3 V
3	SPI6_MISO	3.3 V
4	SPI6_MOSI	3.3 V
5	SPI6_nCS1	3.3 V
6	SPI6_DRDY1	3.3 V
7	SPI6_nRESET	3.3 V
8	GND	GND

PWM EXTRA · 6-PIN JST-SH

PIN	SIGNAL	VOLTAGE
1	PWM5	3.3 V
2	PWM6	3.3 V
3	PWM7	3.3 V
4	PWM8	3.3 V
5	PWM9	3.3 V
6	GND	GND

GPS · 6-PIN JST-GH

PIN	SIGNAL	VOLTAGE
1	5V_OUT	5 V
2	USART1_TX	3.3 V
3	USART1_RX	3.3 V
4	I2C1_SCL	3.3 V
5	I2C1_SDA	3.3 V
6	GND	GND

TELEM · 6-PIN JST-GH

PIN	SIGNAL	VOLTAGE
1	5V_OUT	5 V
2	UART7_TX	3.3 V
3	UART7_RX	3.3 V
4	UART7_CTS	3.3 V
5	UART7_RTS	3.3 V
6	GND	GND

VTX · 6-PIN JST-GH

PIN	SIGNAL	VOLTAGE
1	12V_OUT	12 V
2	GND	GND
3	UART5_TX	3.3 V
4	UART5_RX	3.3 V
5	USART2_RX	3.3 V
6	GND	GND

DEBUG · 6-PIN JST-SH

PIN	SIGNAL	VOLTAGE
1	3V3_OUT	3.3 V
2	USART3_TX	3.3 V
3	USART3_RX	3.3 V
4	SWDIO	3.3 V
5	SWCLK	3.3 V
6	GND	GND

RC · 4-PIN JST-GH

PIN	SIGNAL	VOLTAGE
1	5V_OUT	5 V
2	USART6_RX	3.3 V
3	USART6_TX	3.3 V
4	GND	GND

CAN · 4-PIN JST-GH

PIN	SIGNAL	VOLTAGE
1	5V_OUT	5 V
2	CAN1_P	5 V
3	CAN1_N	5 V
4	GND	GND

POWER AUX · 3-PIN JST-GH

PIN	SIGNAL	VOLTAGE
1	12V_OUT	12 V
2	GND	GND
3	VBAT	5.5–54 V

ARK-DS-ARK-FPV | Version 1.0, 2026-10-29 | Compiled from the ARK FPV Production schematic Rev 4.2 (4 Mar 2026), the ARK FPV STEP model, the PX4 and ArduPilot board definitions, and ARK Electronics published documentation. Temperature ratings per component data sheets (JST connectors and microSD socket −25 °C; MCU, barometer and magnetometer +85 °C). Specifications are subject to change without notice. © 2026 ARK Electronics LLC.